Class Is a collection of Data and Member Functions.

Also Called as Prototype that defines the Variables and Functions Common to all Objects of certain kind. Can be referred as User Defined Data Type or Abstract Data Type (ADT)

Class can also be defined as ***Blue Print*** of an object.

Data Members : Variables holds the Values also called as Data Members

Member Functions : Performs Operations of Data Members also called as Member Functions

Declaration :

Always declared by the Keyword class

Syntax :

```
class class_name
{
    Access Specifier : Declarations → Data Members ( Variables );
    Access Specifier : Operations → Members Functions;;
};
```

Eg :

Student Class :

```
class Student{
    private:
        studentid char[8];
        batchid char[8];
        name char[50];
        email char[255];
```

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```
        phone char[10];
    public:
        void getDetails(){
            cout << “Enter ID, Name, Batch, Email and Phone\n”;
            cin >>studentid >>batachid >>name >>email>>phone;
        };
        void diaplayDetails(){
            cout<<studentid<< batachid<<name<<email<<phone;
        };
}
```

Access Specifier / Access Modifiers :

Are the labels that specify type of access given to members of a class. Also called as visibility modes.

Usage : Data Hiding

Types of Access Specifiers :

Following are the 3 different Access Specifiers defined in C++

Private :

Members which are defined as Private can only be accessed by Member Functions defined in the Class itself. But cannot be accessed by any *Non Members* of the class.

Declaration :

Always declared by the Keyword private

Public :

Members which are defined as Public can only be accessed by Member Functions and Non members of a class .

Declaration :

Always declared by the Keyword public

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Protected :

Members which are defined as Protected whose access lies between public and private. This specifiers is mainly used while implementing the Inheritance Concept

Declaration :

Always declared by the Keyword protected

Note :

1. If no Access Specifier is Defined, by default the members are treated as **private** only.

2. Default Access Specifier in Structure is **public**

3. Default Access Specifier in Class is **private**

Object :

Is an instance of a class. A real world entity that has a state and a behavior

Syntax :

ClassName ObjectName;

Eg :

Student V19HE6Z1;

Student → Student Class

V19HE6Z1 → Object of Class Student

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